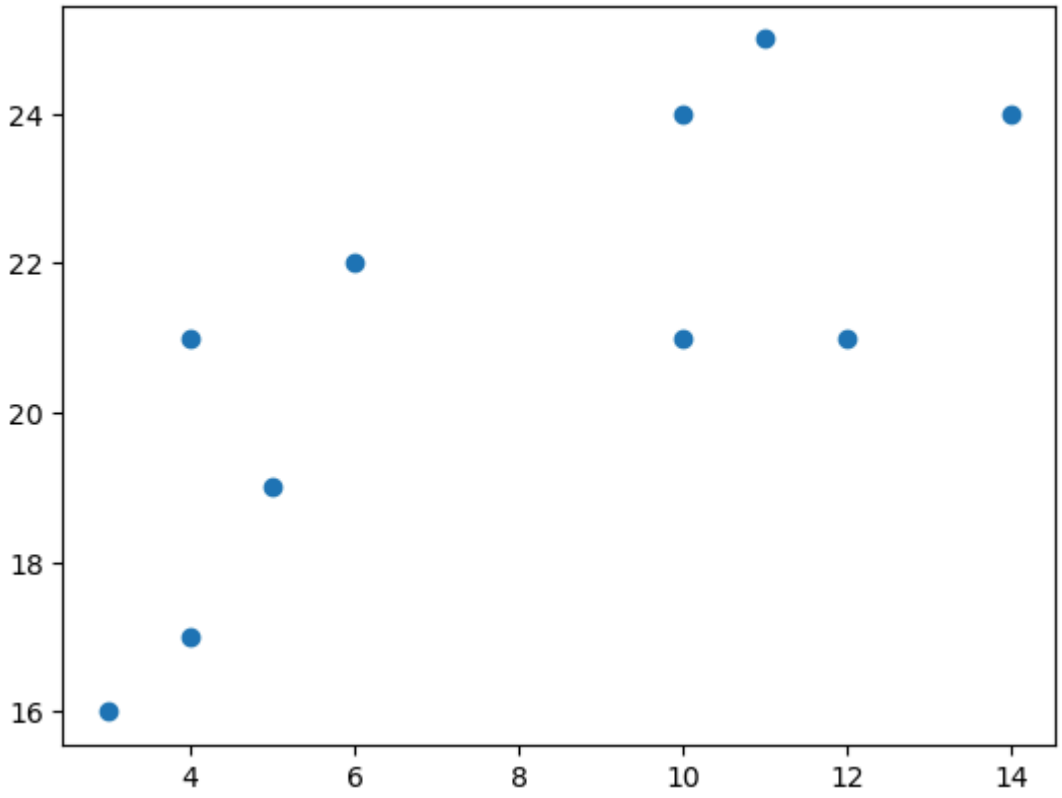
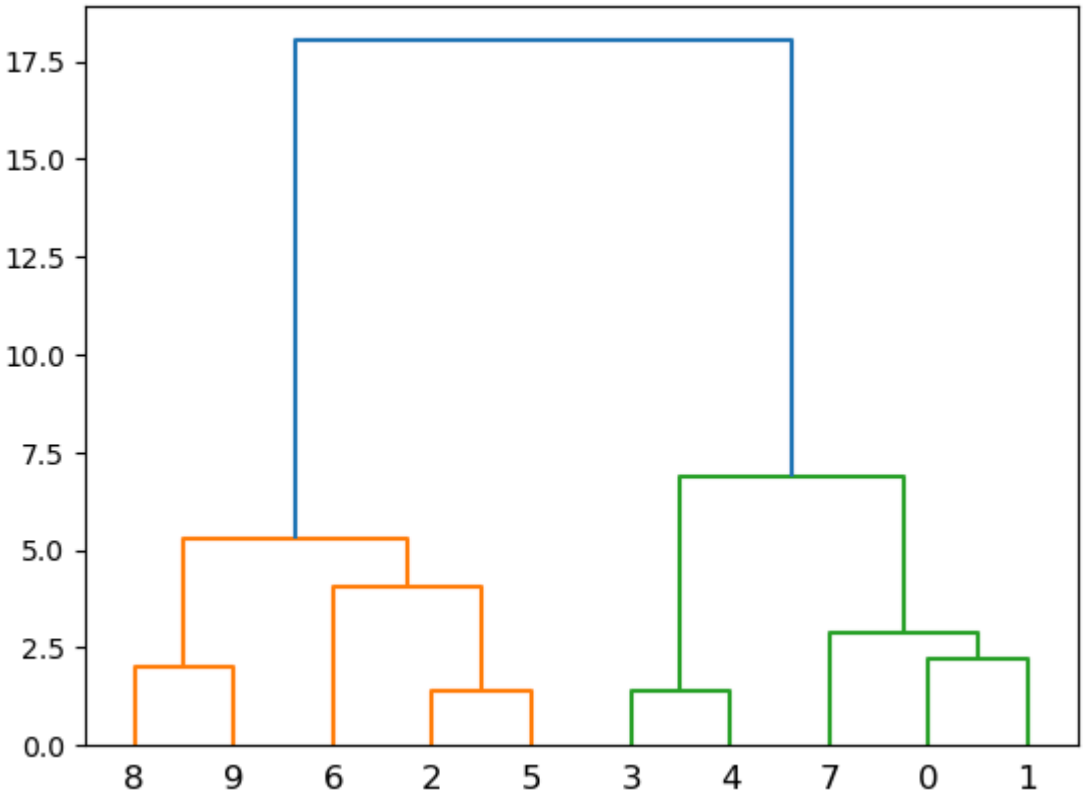
In [2]:
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.cluster import AgglomerativeClustering
```

In [3]:
```
x=[4,5,10,4,3,11,14,6,10,12]
y=[21,19,24,17,16,25,24,22,21,21]
```

In [4]:
```
plt.scatter(x,y)
plt.show()
```



In [5]:
```
data=list(zip(x,y))
linkage_data=linkage(data,method="ward",metric='euclidean')
dendrogram(linkage_data)
plt.show()
```



In [8]:
```
hierarchical_cluster=AgglomerativeClustering(n_clusters=2,metric='euclidean',lin
labels=hierarchical_cluster.fit_predict(data)
print(labels)
```

```
[0 0 1 0 0 1 1 0 1 1]
```

In [ ]: